MAST20030 Differential Equations
— Problem Sheet 4: Fourier Series —

1. Compute the Fourier series of each of the following functions, extended periodically from the domains indicated, and determine the function that it (pointwise) converges to. In each case, is the convergence uniform?

   (a) $f(x) = H(x), \quad -L < x < L$

   (b) $f(x) = \begin{cases} -L - x & \text{if } -L < x < 0, \\ L - x & \text{if } 0 < x < L, \end{cases}$

   (c) $f(x) = \begin{cases} -1 & \text{if } -2 < x < 0, \\ 1 & \text{if } 0 < x < 2, \end{cases}$

   (d) $f(x) = \begin{cases} 0 & \text{if } 1 < |x| < 2, \\ x & \text{if } |x| < 1, \end{cases}$

   (e) $f(x) = \dfrac{\mathrm{e}^{-|x|/\epsilon}}{2\epsilon}, \qquad -1 < x < 1$

   (f) $f(x) = \begin{cases} -\pi - x & \text{if } -\pi < x < -\frac{\pi}{2}, \\ x & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2}, \\ \pi - x & \text{if } \frac{\pi}{2} < x < \pi. \end{cases}$

2. Consider the waveform

$$f(t) = e^t, \quad -\pi < t < \pi$$

   where $f(t) = f(t + 2\pi)$.

   (a) Sketch the graph of $f(t)$ for $-3\pi \le t \le 3\pi$.

   (b) Write down a general expression for the Fourier series of f.

   (c) Does the Fourier series converges uniformly? If so, prove this using the Weierstrass test.

3. A sinusoidal voltage $\sin t$ is passed through a half-wave rectifier to produce a periodic output voltage with period 2π such that

$$E(t) = \begin{cases} 0, & -\pi < t < 0 \\ \sin t, & 0 < t < \pi \end{cases}$$

   (a) Sketch the graph of $E(t)$ for $-3\pi \le t \le 3\pi$.

   (b) Write down a general expression for the Fourier series of E.

   (c) Does the Fourier series converges uniformly? If so, prove this using the Weierstrass test.

4. Consider the function

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ 2, & 1 < t < 2 \end{cases}$$

   (a) Sketch $f_e(t)$, the even periodic extension of f in the range $-4 \leq t \leq 4$.

   (b) Write down a general expression for the Fourier series of f_e.

   (c) What values does the Fourier series for f_e converge to if $t = 0$ and $t = 2$?

   (d) Sketch $f_o(t)$, the odd periodic extension of f in the range $-4 \leq t \leq 4$.

   (e) Write down a general expression for the Fourier series of f_o.

   (f) What values does the Fourier series for f_o converge to if $t = 0$ and $t = 2$?

5. Consider the function

$$f(t) = t^2 \qquad \text{for } 0 < t < \pi$$

   (a) Sketch $f_e(t)$, the even periodic extension of f in the range $-3\pi \leq t \leq 3\pi$.

   (b) Write down a general expression for the Fourier series of f_e.

   (c) Sketch $f_o(t)$, the odd periodic extension of f in the range $-3\pi \leq t \leq 3\pi$.

   (d) Write down a general expression for the Fourier series of f_o.

6. Consider the function

$$f(t) = \begin{cases} -(\pi + t), & -\pi < t < 0 \\ \pi - t, & 0 < t < \pi \end{cases}$$

   (a) Obtain a general Fourier series representation for f, if f is periodic with period 2π.

   (b) Using part (a), find a particular solution for the non-homogenous ordinary differential equation

$$-y'' + y = f$$

   (c) Find the general solution for the differential equation in part (b).

7. Consider the function

$$f(t) = \frac{t^2}{4} \qquad \text{for } -\pi < t < \pi.$$

   (a) Obtain a general Fourier series representation for f, if f is periodic with period 2π.

   (b) Using part (a), find a particular solution to the differential equation

$$y'' + y' + y = f$$

   that describes the oscillations of a damped mechanical system subject to the periodic driving force f.

   (c) Using part (a), evaluate the sum

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

MAST20030 Differential Equations
— Solutions to Problem Sheet 4: Fourier Series —

1. (a) $s(x) = \frac{1}{2} + \frac{2}{\pi}\sum_{m=1}^{\infty} \frac{1}{2m-1} \sin k_{2m-1}x$

   (b) $s(x) = \frac{2L}{\pi}\sum_{n=1}^{\infty} \frac{1}{n} \sin k_n x$

   (c) $s(x) = \frac{4}{\pi}\sum_{m=1}^{\infty} \frac{1}{2m-1} \sin k_{2m-1}x$

   (d) $s(x) = \frac{2}{\pi^2}\sum_{n=1}^{\infty} \frac{2\sin k_n - n\pi \cos k_n}{n^2} \sin k_n x$

   (e) $s(x) = \frac{1}{2}(1 - \mathrm{e}^{-1/\epsilon}) + \sum_{n=1}^{\infty} \frac{1-(-1)^n \mathrm{e}^{-1/\epsilon}}{1+n^2\pi^2\epsilon^2} \cos k_n x$

   (f) $s(x) = -\frac{4}{\pi}\sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \sin k_{2m-1}x$

2. $s(t) = \frac{\sinh \pi}{\pi} + \frac{2\sinh \pi}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^n}{1+n^2}(\cos \omega_n t - n \sin \omega_n t)$

3. $s(t) = \frac{1}{\pi} + \frac{1}{2}\sin t + \frac{2}{\pi}\sum_{m=1}^{\infty} \frac{1}{1-4m^2} \cos 2mt$

4. (b) $s(t) = \frac{3}{2} + \frac{2}{\pi}\sum_{m=1}^{\infty} \frac{(-1)^m}{2m-1} \cos \omega_{2m-1} t$

   (e) $s(t) = \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{1-2(-1)^n + \cos \omega_n}{n} \sin \omega_n t$

5. (b) $s(t) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos \omega_n t$

   (d) $s(t) = -\frac{2}{\pi}\sum_{n=1}^{\infty} \frac{2+(-1)^n(n^2\pi^2-2)}{n^3} \sin \omega_n t$

6. (a) $s(t) = \sum_{n=1}^{\infty} \frac{2}{n} \sin nt$

   (c) $y(t) = C_1 \mathrm{e}^t + C_2 \mathrm{e}^{-t} + \sum_{n=1}^{\infty} \frac{2}{n(1+n^2)} \sin nt$

7. (a) $s(t) = \dfrac{\pi^2}{12} + \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nt$

   (b) $y_p(t) = \dfrac{\pi^2}{12} + \displaystyle\sum_{n=1}^{\infty} \left(\frac{(-1)^n (1-n^2)}{n^2(1-n^2+n^4)} \cos nt + \frac{(-1)^n}{n(1-n^2+n^4)} \sin nt \right)$

   (c) $-\dfrac{\pi^2}{12}$